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Geometric Algebra for Multidimension-Unified Geographical Information System

Linwang Yuan*, Zhaoyuan Yu, Wen Luo, Lin Yi, and
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Abstract. Traditional Euclidean geometry-based Geographical Information System (GIS) is not multidimensional unification with weak ability to object expression and analysis of high dimensions. Geometric algebra (GA) can connect different geometric and algebra systems, and provide rigorous and elegant foundation for expression, modeling and analysis in GIS. This paper proposes the implementation methods for system construction and key components of multidimension-unified GIS. Based on such properties as multidimension-unified and coordinate-free of GA, data models, data indexes, and data analysis algorithms for multidimensional vector data, raster and vector field data are developed. This study indicates that GA provides a new mathematical tool for the development of GIS characterized as multidimension-unified expression and computation. For the development of geographical analysis methods, it can represent multidimensional spatio-temporal changes conveniently.

Keywords. Geometric algebra; multidimension-unified GIS; data model; data index; spatio-temporal analysis.

1. Introduction

With the development of 3D and temporal GIS, high dimensional data analysis ability, computation complexity and extensibility have become the handicap of existing GIS [1]. Object expression and computation based on Euclidean geometry are highly dependent on object coordinates and coordinate systems. The expression and analysis of different dimensional geographical objects are disunity [2]. Spatial analysis algorithms in geometry-based GIS handle the computation of different dimensional objects (e.g. point, line, plane and sphere) respectively, which causes problems of ambiguity of spatial

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meaning, complexity of spatial simulation and inference, etc. [3]. Moreover, as the fusion analysis of geometric objects with different dimensions, it is rather difficult to be linked and integrated effectively with geographical models [4, 5]. Therefore, expansion of spatial analysis algorithms for high dimension space is very complex, and it is also disadvantageous to improve the efficiency of spatial analysis algorithms.

Vector spatio-temporal data is complex and diverse in structure, storage and expression mode, therefore few methods are suitable for multidimensional vector spatio-temporal data analysis [6, 7]. A variety of GIS data models were developed to support expression, modeling and analysis of geographical objects and scenes [8, 9, 10, 11, 12]. The object expression of most existing data models was under Euclidean geometric framework. Due to the coordinate relative and dimensional isolation characteristics of Euclidean geometric, GIS still faces such problems as complex computation structure and simultaneous integration of ambiguous heterogeneous data including scalar quantity, vector and tensor, etc. [13, 14, 15]. The isolation expression of geometric objects, operations (e.g. measurement, topology, location, etc.) of spatial relations and computation increase the difficulty of unified expression and computation and the complexity and non-determinacy of spatial meaning description and spatial inference [16, 17]. Relevant semantic information is isolated when particular geographical objects are expressed [18, 19].

Geometric algebra (GA) is developed from Hamilton quaternion and Grassmann extension algebra. The key fundamental elements of GA are multivector and geometric products with combination of multidimensional objects into a single structure and the notion of orthogonality and collinearity into one operation [20]. Conformal geometric algebra (CGA) on the perspective of covariation, provides several structures such as Grassmann structure, inner products, unified spinor effects, parenthesis system and invariant system to express geometric objects, relations, changes and senses [21, 22, 23]. These structures have shown distinct advantages in geometric data processing and computation [24]. With the help of GA operators, GA can be used for multidimension-unified expression and analysis in a coordinate-free way [25, 26]. The compactness of expression of GA and the multidimension-unified operations are conveyed, which greatly reduces the computation complexity and improves the analysis abilities [20, 26]. It has already become an important tool in mathematical analysis, theoretical physics, geometry, mechanical theorem proving, etc. [27, 28, 29, 30].

Introducing GA to GIS is a prospective way to improve the multidimensional expression and analysis ability of GIS. GA provides a powerful computational system to solve complex geographical modeling problems [31]. In this paper, we introduce GA to explore construction of multidimension-unified GIS from basic theories to implementation methods. This paper is organized as follows: intrinsic correlations between GA and GIS are discussed in section 2, and object expression and computation framework are introduced in section 3. Then, construction framework and key techniques are presented

in section 4 and 5 respectively. Finally, the conclusion and discussions are given in section 6.

2. Intrinsic Correlations between Geometric Algebra and GIS

Research objects of GIS have developed from simple basic geometric objects in Cartesian coordinate system to complex geographical objects of different forms and types involving many coordinate systems [32, 33]. The change of geographical phenomena can be static or dynamic, continuous or discrete, and referred to one, two, three or four dimensions [34, 35, 36]. Algebraic manipulation of multidimensional spatio-temporal data seems to be an ideal way to improve the integration of representation, editing, storage, index, analysis and algorithm of GIS [31, 37]. One of the most significant challenges of proposing one type of algebraic manipulation to multidimensional GIS data is lack of expression ability for complex geometries and relations with clear geometric meaning, as well as easy computational implementations. Another challenge is to express both spatial and temporal dimensions on a more abstract level, which are of unique dimensions without significant changes, and to generate a more powerful integrative expression framework for unified spatio-temporal data storage, editing, updating and analysis.

Unified expression and effective connection between GA and different algebra and geometric systems can be calculated simply by constructing appropriate computational space [22]. “Products” in GA define such spatial operation sets as geometries, measurements, topologies and directions, and have the ability to achieve expression and mutual transformation among various kinds of space (e.g. Euclidean space, Homogeneous space and Conformal space), and thus support unification and expression of various algebra and geometric systems [22, 23]. The ability of geometric computation and expression in GA makes time an equivalent dimension to space in computational space construction, thus the expression and computation of unified time and space are achieved [27, 29]. Therefore, GA provides an ideal mathematical tool for construction, analysis, and integration of classical Euclidean, Inversive Geometry and Projective Geometry, with practical applications to various geographical analysis [38, 39, 40].

Geographical objects are usually characterized as multidimensional features, and span many different spatio-temporal scales. Accordingly, spatio-temporal analysis involving different type objects with different dimensions requires synchronous analysis and treatment of such elements as geometry, topology and attributes, etc. Meanwhile, it concerns different coordinate systems (e.g. plane coordinate system, spherical coordinate system). GA intrinsically mixes the dimensional computation of definite geometric meaning, achieves integrative expression and storage of objects of different dimensions in mathematical structures, and supports expression, modeling and analysis of complex geographical and phenomena [41, 42]. Unification and integration of GA with different spaces simplify expression and transformation of objects

in different coordinate systems, and unify organization, storage and computation of relevant spatial data, thus they provide foundations for developing multidimension-unified GIS analysis algorithms [31, 43, 44]. GIS data models and topological analysis models also can be extended in GA framework, which will help building unified descriptive framework of multidimensional geographical time and space. And present models and algorithms can be inherited to support geographical models and compact integration of GIS analysis and geographical analysis [43].

GA provides the mathematical foundation for real world geographical object expression and analysis under multidimension-unified framework, which can support complex and dynamic geographical analysis [26, 28]. The advantages can be identified as follows: (1) Multidimensional unification and coordinate-free property provide potentials for unified multidimensional expression and computation of complex geographical objects. The GA-based expression of fractal and complex vector fields enhances expression abilities of complex geographical objects [38, 45, 46, 47]; (2) Unified expression and connection of GA with different algebra and geometric systems can be simplified by construction of appropriate computation space; (3) The developments of discrete dynamic system and complex motion expression [34, 48] simplify transformation from real spacetime to geometric algebra space and extend descriptive and simulative abilities of complex motion processes and discrete/continuous dynamic systems; (4) Motion expression in GA is explicit and unified, and direct intrinsic properties of such kinematical parameters as speed and acceleration, as well as their support for smooth interpolation properties are helpful to achieve unified expression of continuous and discrete geographical phenomena [20, 23, 24, 29, 38].

3. Expression and Computation for Multidimension-Unified GIS

Multidimension-unified object expression is the primary foundation of GA-based multidimension-unified GIS. The adaptive expression methods integrate the dimensional structure with geometric relations. The computational model has intrinsic characteristics and clear structures based on GA operators. With the help of the multidimension-unified object expression framework, the concordancy expression of multidimensional objects, the unified space, time, motion and transformation are achieved. The multidimension-unified computation framework is also proposed, which can support complex geo-computation.

3.1. Space Construction and GIS Computation

Computational space construction and geometric object expression based on geometric products in GA include directly unified computation of different dimensions on the basic computational aspects. Four computational spaces, the Euclidean space, homogeneous space, conformal space, and spacetime algebra

space, are used to achieve the expression of geometric objects with simultaneous inclusion of dimensional construction relations and metric relations. Despite of the spacetime algebra space, Euclidean space embedded in projective or conformal space is directly corresponding to Cartesian space. The unification of CGA to Euclidean geometry, projective geometry, hyperbolic geometry and spherical geometry can respectively correspond to different invariant changes based on null-vectors [20, 22, 26]. In fact, homogeneous and conformal spaces convey projective space and spherical space, which is very frequently used in geographical analysis. The coordinate-free characteristics of CGA also help to handle these problems naturally and directly, with the connotation and intuitive geometric interpretation. Therefore, GA can be considered as a unified descriptive language to connect algebra with geometry, mathematics with physics, abstract with concrete spacetime [27, 28].

Distinct from those geometric operations based on Euclidean geometry, in the GA framework, both geometry and geometric relations are based on the algebraic representation of the geometric products. The hierarchical Grassmann structure is corresponding to the hierarchical structure of the geometry. Different dimensional geometric objects constructed by the outer product can directly support calculation. The operators in GA can be extended to concise and intuitive spatial analysis operations, and provide foundation for multidimension-unified operation of geometric objects. Clear geometric meaning can be retained in the representation of objects. The objects expression and their computation parameters are directly contained in the representation of operations so that adaptable reconstruction can be performed during a computational procedure. Its representation-intrinsic properties, namely the metric property and calculable property, make the representation and analysis of complex geographical phenomena easier. And multidimension-unified operations are extended to arbitrary object representation as well as relational calculus. By defining a universal storage structure, representational and operational structures can be submitted to a more sophisticated but efficient analysis [31, 43, 44]. Utilizing the unification and connection of GA with different algebra and geometric systems, most existing algorithms or operations can be simplified by construction of appropriate computation space.

3.2. Multidimension-unified Object Expression

By using inner, outer and geometric products, objects are expressed including dimensional hierarchy and geometric relations. Different kinds of algebra transformation and reflection are then proposed. Grassmann structures of CGA are used for multidimensional geographical objects expression. Basic multidimensional geometric objects such as points, planes etc. are all formulated in an outermorphism way. The inner products are also used to inherit geometric features and relationships from the construction of the objects, and intrinsic metric properties are also integrated in the multivector expression [43, 44]. Metric parameters such as intrinsic distance or angles are then automatically incorporated in the parametric expression for each geometric object

[49]. Based on the multivector structure of uniform expression of variety of objects in GA space, methods of construction, resolution and expression of multidimensional objects are developed to achieve modeling and storage of geographic objects [44].

Expressing and modeling of geometric objects including geometric hierarchy and metric relations can be effectively simplified and calculated with multivector structures [43]. Similar to simplex-complex structures in GIS, multidimensional geometric objects with different structures can be decomposed into sets of single geometric elements (e.g. points, lines, planes, spheres, etc.) of different dimensions. Even the most complicated feature can be subdivided into several simple simplexes. After transforming these simplexes into GA space, they can then be modeled and stored by blades, and linked with the multivector structures [43, 50]. Effective combination of the outer and inner products equations of CGA makes objects able to simultaneously integrate hierarchical geometry relationships, coordinates and measurements during its expression and modeling procedure. This will achieve integrative expression of objects of different dimensions, quick extraction, regrouping and effective analysis of objects of different dimensions based on multivector. Thus the support of the unified organization, storage of geographical objects and geographical scenes of different dimensions can be achieved.

Proper Euclidean motions can be represented by an even number of reflections, and therefore by a versor consisting of an even number of vector terms (appropriately called an even versor). The general rigid body motion in Euclidean space can be expressed as $O_r = TRSoT'R'S'$, where T is the translation operator, R is the rotation operator, and S is the scaling operator [20]. Thus, transformations such as reflection, rotation, translation, scaling or their combinations can be unified with the sandwich model ($X \rightarrow X' : X' = V XV^{-1}$). The transformation operator V , called the versor, is generalized in CGA. All the versor motion expression preserves Euclidean distances, though not necessarily handedness [20, 24, 29]. These transformations are continuously connected to the identity, and can be analyzed more reliable and accurate. Translation, rotation, and scaling transformations in CGA framework can be unified into a single versor, which can also be expressed exponentially for smooth interpolation. Motion expression in GA is explicit and unified, and direct intrinsic properties of such kinematical parameters as speed and acceleration, etc. as well as their support for smooth interpolation properties are helpful to achieve unified expression of continuous and discrete geographical phenomena [21, 22, 29, 30].

3.3. Multidimension-unified Computation

Property-embedded, parameterized expression and calculable property based on the expression of geometric objects and the geometric relations of geometric products make multidimension-unified computation possible. Adaptive and self-contained properties of basic computation and transformation

operators of GA make computation results only connecting with geometric properties of objects themselves, rather than connecting with their dimensions, coordinates and expression forms. Computational space construction and geometric object expression based on inner and outer products in GA include directly unified computation of different dimensions. Such characteristics as multidimensional unification and coordinate-free property of GA computation bestow object expression and spatial relation computation based on GA characterized as property-intrinsic parameterized and adaptive [21, 22, 30], and support directly parallel computation from the bottom. Different transformation in GA framework will keep geometric meaning and properties of original geometric objects after geometric spatial transformation. When transformation expression meets covariant conditions, it is possible to construct adaptive dynamic computation of spatial relations with unified structure and definite meaning [21, 22].

Object expression and scene organization with multivectors can directly facilitate computation. Parameterization expression and dimensional adaptability ensure simplicity and commonality of algorithms [20, 43]. The GA operators are computational stackable, order-independent and structure-preserving, which can be used to establish an adaptable representation, and to execute dynamic reconstruction of geometric objects. Structural characteristic analysis of transformation operator components, state parameters, such as velocity, angle, rotational axis, etc., can be performed directly. An effective unification of representational and operational structure can be implemented to develop various spatial analysis algorithms [31, 43, 44, 51, 52, 53]. Therefore, algorithms can be constructed to achieve sophisticated spatial analysis with concordant output. By introducing advanced technologies such as computational template construction and pre-compiled and hardware accelerating with GPU and FPGA, most GA algorithms have nearly or even much better efficiency than traditional algorithms with vector algebra or matrix algebra [54, 55, 56]. Therefore, GA-based spatial relation analysis and computation provide new methods for spatial relation analysis under complex and dynamical environments.

Expression structure based on multivector makes geometric objects and their relational expression multidimension-unified and highly independent. Operators and algorithms for GIS computation and analysis to support analysis methods of geographical models are then constructed. Basic GA operators are used in combination with geometric transformation and reflection and expression of geometric relations to construct GA operator libraries. Equation solving of geometric relations, Clifford convolution and Clifford FFT, spacetime algebra method and versor interpolation, etc. are then developed [44, 45, 46, 51, 52, 53]. GIS algorithms and geographical analysis models (e.g. dynamic scene simulation, motion expression and interpolation, network analysis models, characteristic analysis of spatio-temporal fields, change detection and curvature computation, etc.), are developed to realize GIS analysis and geographical computation.

4. The Constructive Structure of Multidimension-Unified GIS

Founded on the multidimension-unified expression and computation framework, we propose the layer structure of multidimension-unified GIS based on GA. The three-layer holistic framework integrating theoretical concepts, functional structures and application structures are given firstly. Then the construction and implementation framework composed according to the data flow, including data input, object analysis, GA computation, resulting analysis, data output and visualization are discussed. And the design of GA computational engine is also presented.

4.1. Holistic Framework of Multidimension-unified GIS based on GA

The holistic framework of multidimension-unified GIS based on GA consists of theoretical framework, system implementation and application demonstration (see Figure 1). According to the temporal/spatial data organization ways of different types and organizational forms and characteristics of common GIS data, space transformation between geographical space and GA space is constructed with advantages in dimensional changes and relation computation. It provides space transformation strategies and mechanisms from vector and raster data to GA spaces such as Euclidean space, homogeneous space, and conformal space, etc. Basic computation of inner, outer and geometric products is constructed in relevant computational space for transformation between different coordinate systems and characteristics, in order to provide theoretical foundation and computation space for GIS analysis. Then, multidimension-unified data models, relevant data storage and index mechanism based on GA are constructed. GIS operators, such as spatio-temporal topological, transformational and analysis operators, are defined to extend basic GA operators and construct mapping functions to define GA algorithms for geographical analysis and GIS analysis. Then GA computation engine is constructed to implement spatial analysis operator libraries. As to the application demonstration, analysis archetypal system of multidimension-unified GIS with strong openness and extensibility is constructed on the basis of software engineering [43, 57]. Pleophyletic data is integrated with construction and application of such typical algorithm as emergency evacuation, effect evaluation, property match and evolutionary simulation, etc.

4.2. The Constructive Framework of Multidimension-unified GIS

The constructive framework of analysis algorithms in multidimension-unified GIS is composed of basic calculus and basic operators, geometric measurement and topological analysis, property analysis, motion expression, GIS algorithms and geographical models, etc. (Figure 2). Computational space construction based on GA as well as operational rules and computational properties applicable for geographical analysis and computation are defined and implemented in basic operations of GA, such as geometric reverse, geometric opposition, etc., which are introduced to construct basic operators of CA such as projection and reflection, etc. Inner and outer products expressions of geometric objects are combined to construct such geometric measure operators

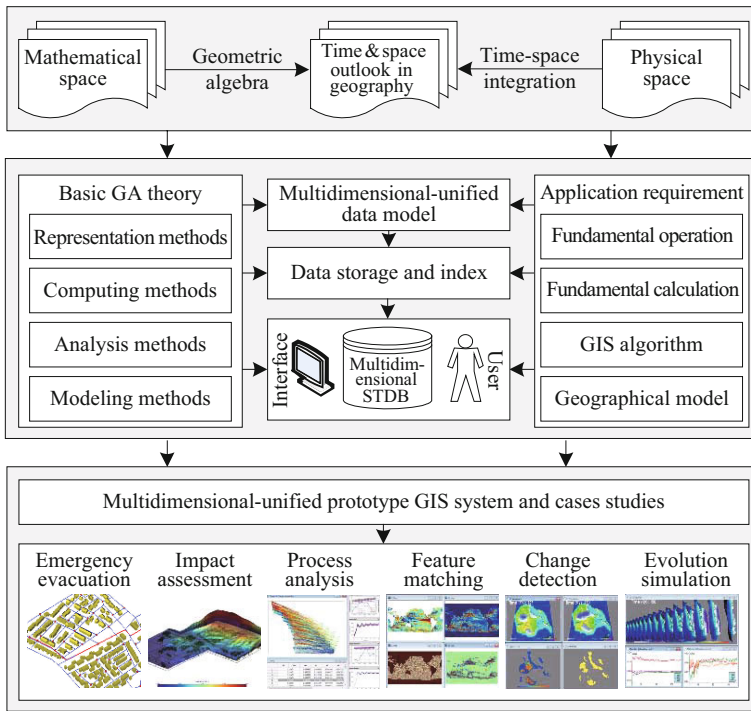


FIGURE 1. The diagram of Holistic framework

as distance, area, etc., and spatial relation operators of geometric objects. Motion expression operators expressing real geographical dynamic evolutionary processes are built to achieve forms and spatial relations of geometric objects in unified spatio-temporal framework and efficient computation and accurate expression of motion process. And then geographical models of GIS spatial analysis with topological analysis to achieve encapsulation and integration of operators and algorithms are designed.

Multidimension-unified GIS computation and analysis methods are implemented with GA operators and algorithms. Space construction, mutual transformation in different GA space and basic operational definition and operator construction methods in different spaces are developed. Quick regrouping of multivector, data structures and operational structures, analysis and operations are defined. Corresponding analysis methods of multivector and analysis types, meanings, and characteristic parameters of objects, such as Blade, Versor, etc. are constructed to achieve quick analysis, retraction, regrouping and computation of geographical objects and geographical scenes based on multivector expression. GIS operational procedure based on different types of GA operators are relevance of spatial operations. Dynamic regenerative needs of computational processes and results are considered.

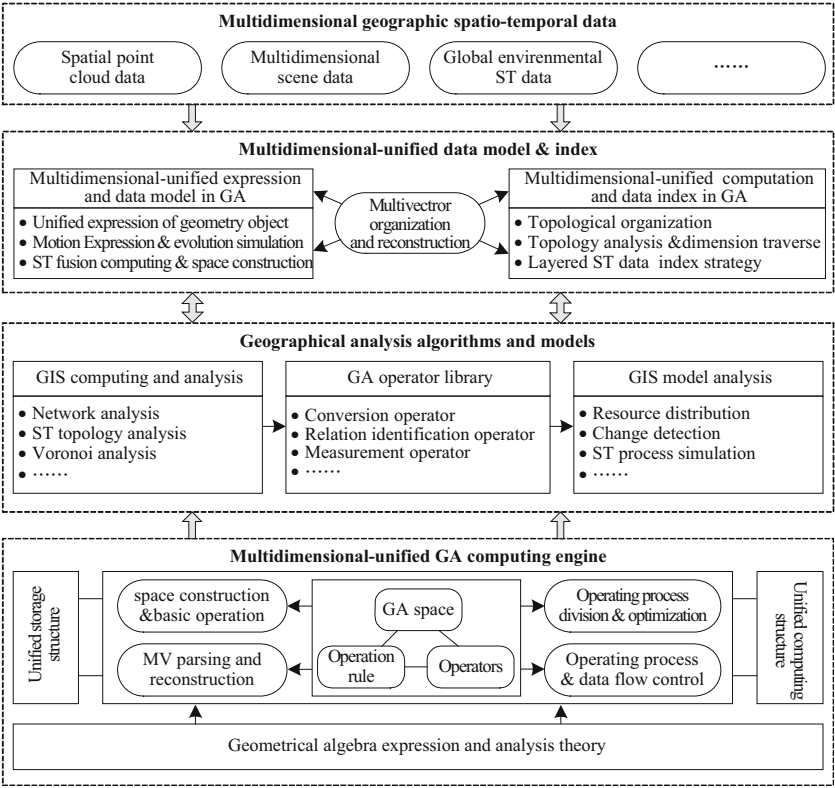


FIGURE 2. The constructive framework of multidimension-unified GIS

And the computing processes can be optimized by the use of such techniques as preprocessing tables and grade-tracking. Formalized expressions and computational models of geometric objects and spatial relations are constructed. An integrative data flow mode of multidimensional data, metadata, parametric data and information data to build process control and data flow management of operations to support complex spatial computation are then proposed.

Computation engines of GA facing multidimensional GIS analysis are constructed to achieve systematic integration by the use of plug-in framework [31, 58]. Founded on the library of GDAL/ORG and specific GIS data format adapters, the data engine which can extract basic geometric and topological information as well as the sematic and attribute data under the premise of keeping the compatibilities with the commonly used GIS data. Focusing on the computation tasks, the operator library, which organizes the basic and hybrid Clifford operators, is designed with several GA computational engines (Gaigen2, Clu and Gaalop [26, 54, 55]). Algorithm libraries focused on specific geographical analysis, such as network analysis, Voronoi diagram, or

spatial-temporal dynamical data analysis are then constructed. Other computational models of classic geographical analysis methods are also integrated. By abstracting the above functions and methods, we form the computational Application Programming Interface (API), which has been used as the foundation of the CAUSTA system [31]. ‘Pluggings’, providing the functions to solve certain types of geographical analysis problems, are then designed in the CAUSTA system. The user interaction and visualizations are also directly implemented in the CAUSTA system.

5. Implementation of Key Components

In order to form a full functional GIS, we fuse our thought of constructing multidimension-unified GIS with GA and design the key components of a real applicable GIS system. The construction structure and design of multidimensional data model, spatial index for solid objects and typical kinds of GIS analysis algorithms with data types of vector, raster, vector fields and networks are proposed. Demonstration with real world data, the user interface and visualization are also presented.

5.1. Multidimension-unified GIS Data Model

Expression, organization and modeling of geographical objects of different dimensions are mainly achieved in construction of data models [43]. Blades are used to express every geometric feature class, geometric algebra expressions of complex geometric objects which are achieved with multivectors linking the representation of the above geometric feature classes. The mechanism of abstraction and spatial transformation of objects and algebra space construction are firstly implemented. Inner products, outer products and GA operators are used to express geometric objects based on mathematical equations and geometric transformations. Boundaries can be set by choosing the right sequence of points according to some criterion. Since the dimension of geometric objects, which are expressed as blades, are consistent with their Grassmann structure, the basic geometric objects need only point sets of corresponding numbers with Grassmann levels to create an actual representation. For convenience in managing scene objects and performing spatial analysis, the multivector representation of the scene objects is organized according to types, points, lines, and planes as expressed by the outer products that are arranged according to the grade (Figure 3).

The multivector construction provides the fundamental basis for the data storage structures and implementation of design strategies, and structure and work flow optimization. Perpendicularity of the blade basis can be fulfilled for objects of different dimensions and can be calculated by appropriate choice of vectors. The multivector representation of geographical objects can yield their geometric components. The Grassmann structure consists with the dimensional structure [57]. Even the most complex objects can be adaptively expressed and organized with the above structures (Figure 3). The expression is also structure adaptive and computable. Then we can calculate

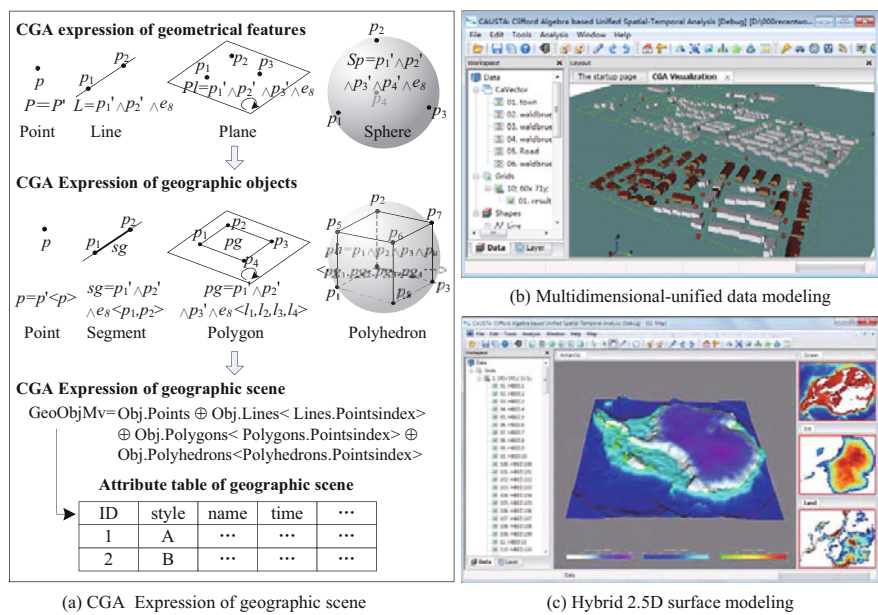


FIGURE 3. Multidimension-unified GIS Data Modeling

the geometric and measurement relations of their own geometric components and of different geographical objects. In addition, most operators and algorithms associated with spatial analysis can be employed to a complex shape sets, thus the operators can be applied together.

5.2. Multidimension-unified Index Methods

Multidimension-unified indexes are constructed with orientation of multi-dimensional vector data (see Figure 4). The element subdivision of multi-dimensional object voxelization and the boundary restricted non-overlapping sphere filling algorithm is developed to balance the conditions of object voxelization. Taking advantage of simplicity of sphere packing and convenience of geometric relational operations in GA, the spatial optimized splitting is implemented through spatial subdivision and demarcation as the pre-processing, which will support the construction of index mechanism of vector solid objects. Generating and updating minimum boundary sphere and a clustering algorithm of filling, subdivision and hierarchy of non-intersecting discrete balls with boundary limits of multidimensional real objects in 3D space are constructed to construct index hierarchy of solid objects [59]. The ordered hierarchical index structures are built by the use of corresponding clustering and reconstruction. And then index rules and retrieval strategies of search regions are constructed according to index goals to improve retrieval and operation efficiency. The experiments on real data suggest that the proposed index can effectively query any position or region on and in the solid objects,

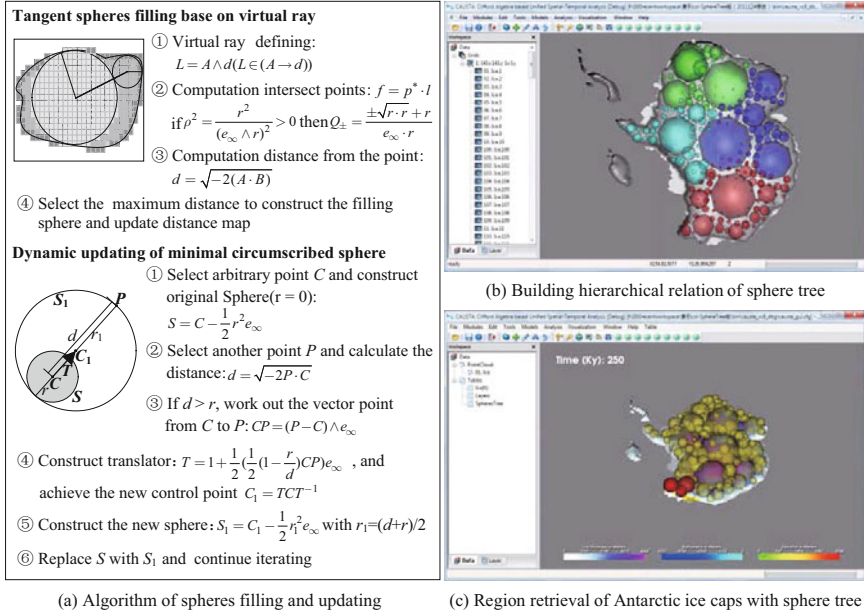


FIGURE 4. Multidimension-unified index methods

and support the nearest linkage distance and dynamical overlapping query under limited time restrictions with high precision [59].

5.3. The Vector Data Analysis Methods

Multidimensional vector data analysis methods, such as multidimensional distance measurements, topology analysis, etc. are designed and extended. Typical examples that need various kinds of multidimensional vector data analysis are the multidimensional unified Delaunay triangulation and the Voronoi diagram. The algorithms are constructed based on CGA. The multidimensional point cloud data are firstly lifted to a higher dimensional space with conformal transformation. The intersection and the dual of convex hull are computed with meet and dual operators. After projection back to the original space, the multidimensional Delaunay triangulation and the Voronoi diagram are constructed. Data structures such as templates of multivectors are designed [60]. Different kinds of geometric objects (e.g. points, point-pairs, lines, rays, hyper-plane triangles, and tetrahedrons) are stored integrally into the data objects, which can be directly calculated with geometric products and operators. Compared with the algorithms commonly used in GIS analysis, performance efficiency, flexibility, and extendibility of data structures are improved especially for high dimensional data.

A multidimensional unified local structure preserving spatial interpolation algorithm was designed based on the Voronoi spatial structure. The

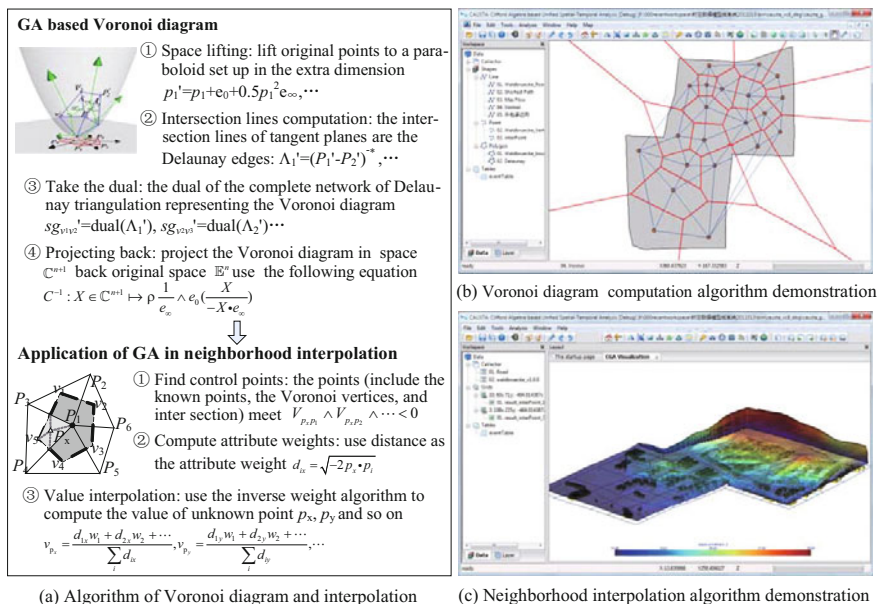


FIGURE 5. Multidimension-unified vector data analysis

multidimensional overlay analysis is designed on the foundation of meet products. By the overlay analysis of the multidimensional unified Delaunay triangulation and Voronoi, the structure characteristic control points for interpolation are generated. The spatial relations such as orientations, distances and contains of control points as well as the original components of Delaunay triangulation and Voronoi are analyzed. Different types of interpolation control points and the processing of their boundary conditions are discussed. Distance of the control points and interpolation points is computed with the inner product, and then the rule of local interpolation is constructed based on inverse distance weighted interpolation to achieve the interpolation algorithm with global and local structural characteristics. The proposed interpolation algorithm can support irregular boundary data, and has well local structure preserving characteristics as well as high accuracy (Figure 5).

5.4. The Raster and Vector Fields Data Analysis Methods

Unified expression of 4D spatio-temporal field of GA is introduced. Splitting, incorporation, regrouping and transformation of multidimensional spatio-temporal cubes methods are firstly constructed with the use of expanding and contracting operators of multidimensional tensor. Indexes of multidimensional spatial data and attributes and analysis data are constructed respectively and connected through concordance lists. Dimension-reduction treatments of original spatio-temporal cubes are done by introducing double complex plane ideas. And the hierarchical spatial index of low dimensions

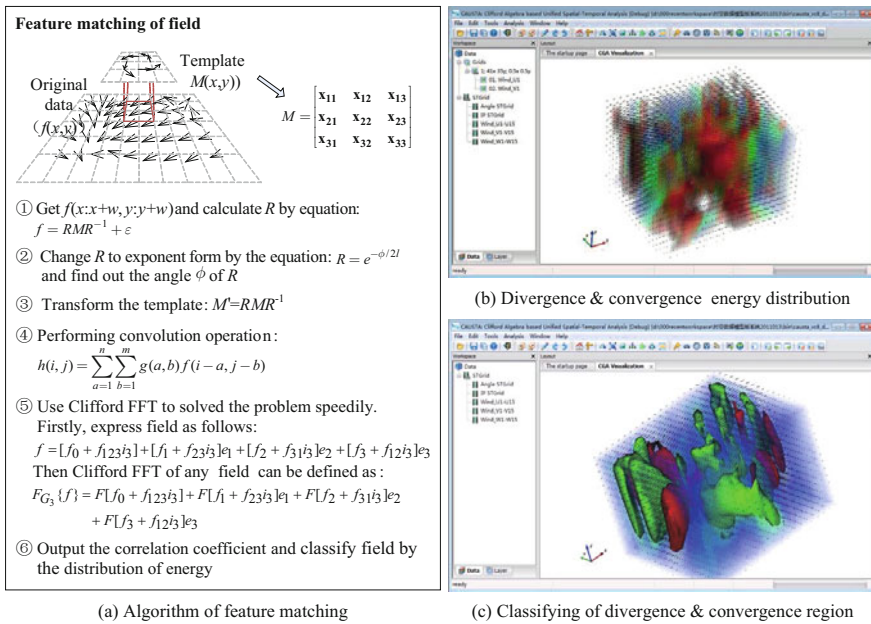


FIGURE 6. Spatio-temporal raster and vector field data analysis

in spatio-temporal cubes or planes after dimension reduction can be implemented, and then form layered dynamic index of original spatio-temporal cubes to achieve efficient quick retrieval of spatio-temporal field data of high dimensions [31].

The multivector expression and computation of vector field data are constructed with the unified expression of geometric products. The vector differentiations and the unified calculation of divergence and curl are computed by template convolution. By using the multidimensional unified and simplicity expression of movement characteristics of geometric algebra, the multidimensional vector field structure adaptive template matching method is proposed. The optimal rotor between the original vector field and the standard template is established based on SVD. The data adaptive divergence-convergence template generation method is then constructed based on the structure consistency of rotor rotation, and the classification of geometric structure of the vector field based on the rotor rotation angle is proposed. With the Clifford Fourier Transformation, the rapid multidimensional vector field convolution, the vector field similarity measurements and the structure characteristic matching methods are designed (see Figure 6). These methods can not only reproduce the geometric associations of vector field divergence and rotation parameters as well as their differential geometry characteristics, but also achieve unified and multidimensional coordinate-free calculations, which can also extract signals from continuous vector fields.

5.5. Network Analysis Methods

Focusing on the leakage of integrating multiple type of constrains in existing optimal path algorithms, GA is introduced as a theoretical foundation for network topology expression and algorithm construction. The unified geometric algebra algorithm for solving multi-type constrained optimal path is proposed with the GA coding of networks. The geometric algebra coding and unified expression of networks are firstly proposed. Network nodes are coded with Clifford algebra basis, and the edges and k-walk routes can be expressed by bivectors and k-blades respectively. With the integrated expression of the nodes, edges, paths and the weights, the route extension, emulation and selection are implemented by the outer product applied to the geometric algebra adjacent matrix. Topologies among nodes, edges and routes of networks can be directly calculated, and the network routes can be extended and traversed with outer products. Then topologies among nodes, edges and routes of networks can be directly calculated, and the network routes can be extended and traversed. The computation of network topology and the path weight are synchronously exported during the route extension. Then the network analysis algorithms can be constructed by route filtering and selection. The classical shortest path and the multi-constrained optimal path algorithm are developed to demonstrate the algorithm design framework (see Figure 7). The results suggest that our algorithm is vivid and clear in geometric meaning and, has great advantages on temporal and spatial complexity, which can provide theoretical and methodological support for complex GIS network analysis.

6. Conclusion and Discussion

Complication and high dimension of objects require that GIS should have the ability to support expression and analysis of complex geographical objects and continuous geographical phenomena. Defects of unified expression of multidimensional objects and spatio-temporal integrated analysis in the present GIS make it difficult to support integrative expression, modeling and computation of complex geographical objects and continuous geographical phenomena. In this paper, GA is introduced to explore construction of multidimension-unified GIS from basic theories to implementation methods. The spatio-temporal data models, spatio-temporal index and algorithm constructive framework are proposed, and then such GIS basis functions as unified modeling, scene organization, data operation and retrieval of multidimensional objects are obtained. Analysis methods such as geometric analysis, topological analysis, neighborhood analysis, network analysis, and spatio-temporal analysis are finally constructed and implemented. The study shows that GA is equipped with possible development of multidimension-unified GIS, and it is suitable for building basic framework and application platform of integrative data models of data expression—data analysis—field application.

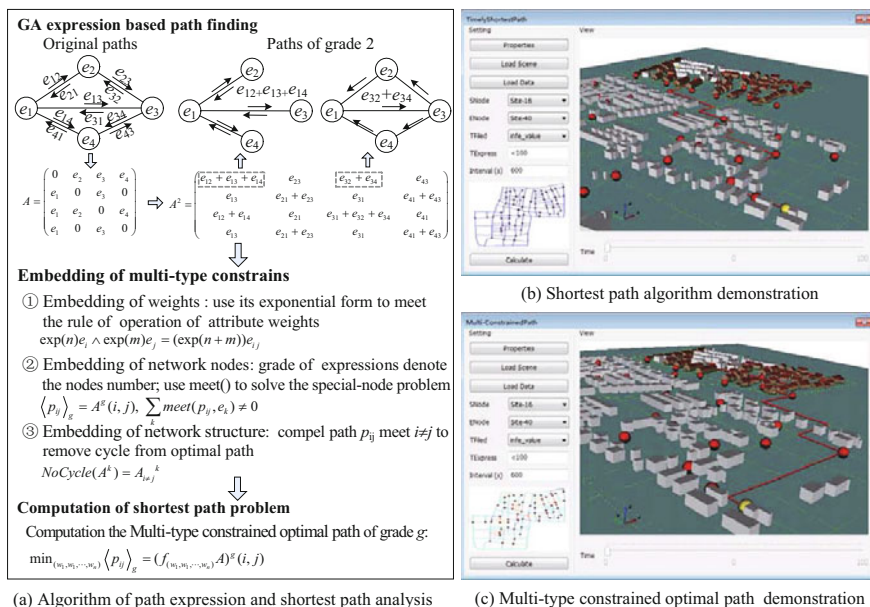


FIGURE 7. GIS network analysis

The GA-based multidimension-unified GIS is still preliminary on current stage. Further work will focus on the following problems: (1) Extending multidimension-unified spatio-temporal data models to support expression and analysis computation of geographical environments. Spatio-temporal data organization, retrieval mechanism and data dynamic dispatching methods in spatio-temporal unified framework can be explored to study semantic structures of objects and spatio-temporal topological properties in spatio-temporal unified framework; (2) Utilizing the properties of different GA computational spaces, analysis methods of spatio-temporal data properties of high dimensions with spatio-temporal dimensional integrative properties should be constructed to solve irregular boundaries and different coordinate systems. Limitations in the present analysis methods of spatio-temporal data properties of high dimensions may be solved to support expression, modeling and simulation of developmental evolutionary processes of geographical objects; (3) Intrinsic support of GA for parallel computation is used to build application algorithms supporting large-scale complex scenes and mass spatio-temporal data analysis to promote present spatio-temporal analysis methods effectively, especially real-time and computation efficiency of vector analysis methods; (4) Spatial analysis platform with high applicability, robustness and extension is developed to promote practical procedure of GA-based multidimension-unified GIS.

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References

- [1] X. Li, J.Q. He, X.Q. Liu, *Intelligent GIS for solving high-dimensional site selection problems using ant colony optimization techniques*. International Journal of Geographical Information Science **23**(4) (2009), 399-416.
- [2] R. Devillers, Y. Bedard, R. Jeansoulin, *Multidimensional management of geospatial data quality information for its dynamic use within GIS*. Photogrammetric Engineering and Remote Sensing **71**(2) (2005), 205-215.
- [3] G. Gerhard, P. Lutz, *How to achieve consistency for 3D city models*. Geoinformatica **15**(1) (2011), 137-165.
- [4] S.W. Wang, Y. Liu, *TeraGrid GIScience Gateway: Bridging cyberinfrastructure and GIScience*. International Journal of Geographical Information Science **23**(5) (2009), 631-656.
- [5] D. Karssenbergh, D.J. KE, *Dynamic environmental modelling in GIS: 1.Modelling in three spatial dimensions*. International Journal of Geographical Information Science **19** (5) (2005), 559-579.
- [6] G. Gröger, L. Plüer, *Topology of surfaces modelling bridges and tunnels in 3D-GIS*. Computers. Environment and Urban Systems **35**(3) (2011), 208-216.
- [7] M. Breunig, S. Zlatanova, *3D geo-database research: Retrospective and future directions*. Computers & Geosciences **37**(7) (2011), 791-803.
- [8] M. Worboys, M. Duckham, *Monitoring qualitative spatiotemporal change for geosensor networks*. International Journal of Geographical Information Science **20** (10) (2006), 1087-1108.
- [9] J. McIntosh, M. Yuan, *A framework to enhance semantic flexibility for analysis of distributed phenomena*. International Journal of Geographic Information Science **19**(10) (2005), 999-1018.
- [10] M. Pilout, K. Tempfli, M. Molenaar, *A tetrahedron - based 3D vector data model for geoinformation*. Advanced Geographic Data Modelling (40) (1994), 129-140.
- [11] D.Y. Shen, K. Takara, Y. Tachikawa, *3D simulation of soft geo-objects*. International Journal of Geographical Information Science **20**(3) (2006), 261-271.
- [12] L. Wu, *Topological relations embodied in a generalized tri-prism (GTP) model for a 3D geoscience modeling system*. Computers & Geosciences **30**(4) (2004), 405-418.
- [13] K.F. Liu, W.Z. Shi, *Computing the fuzzy topological relations of spatial objects based on induced fuzzy topology*. International Journal of Geographical Information Science **20**(8) (2006), 857-883.

- [14] J. Martinez-Llario, J.H. Weber-Jahnke, E. Coll, *Improving dissolve spatial operations in a simple feature model*. Advances in Engineering Software **40** (3) (2009), 170-175.
- [15] M.J. Mineter, *A software framework to create vector-topology in parallel GIS operations*. International Journal of Geographical Information Science **17** (3) (2003), 203-222.
- [16] S. Loch-Dehbi, L. Plüer, *Automatic reasoning for geometric constraints in 3D city models with uncertain observations*. ISPRS Journal of Photogrammetry and Remote Sensing **66**(2) (2011), 177-187.
- [17] R.F. Gholam, U.F. Andrew, S.M. Mohammad, et al., *An ontological structure for semantic interoperability of GIS and environmental modeling*. International Journal of Applied Earth Observation and Geoinformation **10**(3) (2008), 342-357.
- [18] A.H. Monahan, J.C. Fyfe, M.H. Ambaum, et al., *Empirical orthogonal functions: The medium is the message*. Journal of Climate **22**(24) (2009), 6501-6514.
- [19] A. Hannachi, S. Unkel, N.T. Trendafilov, et al., *Independent component analysis of climate data: A new look at EOF rotation*. Journal of Climate **22**(11) (2009), 2797-2812.
- [20] L. Dorst, D. Fontijne, S. Mann, *Geometric algebra for computer science: an object-oriented approach to geometry*. The Morgan Kaufmann Series in Computer Graphics. Morgan Kaufmann Publishers, San Francisco, 2007.
- [21] H.B. Li, *Invariant algebras and geometric reasoning*. World Scientific Publishing, Hongkong, 2008.
- [22] A. Lasenby, *Recent applications of conformal geometric algebra*. In H.Li, P.J.Olver and G.Sommer (eds.). IWMM 2004, LNCS 3519 Springer-Verlag, 2005.
- [23] E. Bayro-Corrochano, *Geometric computing for wavelet transforms. robot vision, control, learning control and action*, Springer Verlag, London. 2010.
- [24] J. Lasenby, W.J. Fitzgerald, A.N. Lasenby, et al., *New geometric methods for computer vision: An application to structure and motion estimation*. International Journal of Computer Vision, **26**(3) (1998), 191-213.
- [25] D. Hestenes, G. Sobczyk, *Clifford algebra to geometric calculus: a unified language for mathematics and physics*. Dordrecht: D. Reidel Publishing, 1984.
- [26] C. Perwass, *Geometric algebra with applications in engineering*. Springer-Verlag, Heidelberg, 2009.
- [27] D. Hestenes, *New foundations for classical mechanics*. Kluwer, New York, 2002.
- [28] M. Pavšič, *Clifford space as a generalization of spacetime: Prospects for QFT of point particles and strings*. Foundations of Physics **35**(9) (2005), 1617-1642.
- [29] D. Hestenes, *Curvature calculations with spacetime algebra*. International Journal of Theoretical Physics **25**(6) (1986), 581-588.
- [30] C. Doran, A. Lasenby, *Geometric algebra for physicists*. Cambridge University Press, New York, 2003.
- [31] L.W. Yuan, Z.Y. Yu, S.F. Chen, et al., *CAUSTA: Clifford algebra based unified spatio-Temporal analysis*. Transactions in GIS, **14**(s1) (2010), 59-83.

- [32] C. Gold, M.A. Mostafavi, *Towards the global GIS*. ISPRS Journal of Photogrammetry and Remote Sensing, **55**(3) (2000), 150-163.
- [33] R.O.C. Tse, C.M. Gold, *TIN meets CAD—extending the TIN concept in GIS*. Future Generation Computer Systems, **20**(7) (2004), 1171-1184.
- [34] M.P. Kwan, J.Y. Lee, *Emergency response after 9/11: the potential of real-time 3D GIS for quick emergency response in micro-spatial environments*. Computers, Environment and Urban Systems, **29**(2) (2005), 93-113.
- [35] Y.C. Lee, M. Molenaar, *Theme issue on dynamic and multidimensional GIS*. ISPRS Journal of Photogrammetry and Remote Sensing **55**(3) (2000), 137-138.
- [36] H. Demirel, *A dynamic multidimensional conceptual data model for transportation applications*. ISPRS Journal of Photogrammetry and Remote Sensing **58**(5-6) (2004), 301-314.
- [37] J. Mennis, *Multidimensional map algebra: Design and implementation of a spatio-temporal GIS processing language*. Transactions in GIS **14**(1) (2010), 1-21.
- [38] E. Bayro-Corrochano, J. Rivera-Rovelo. *The use of geometric algebra for 3D modeling and registration of medical data*. Journal of Mathematical Imaging and Vision **24**(1) (2009), 34:48-60
- [39] D. Hestenes, *New tools for computational geometry and rejuvenation of screw theory*. In Bayro-Corrochano, E.J. and Scheuermann, G. (eds.), Geometric Algebra Computing: In Engineering and Computer Science. Springer, Berlin, (2010), 3–35.
- [40] E. Bayro-Corrochano, L. Reyes-Lozano, J. Zamora-Esquivel, *Conformal geometric algebra for robotic vision*. Journal of Mathematical Imaging and Vision **24**(1) (2006), 24, 55-81.
- [41] E. Bayro-Corrochano, G. Sommer, *Object modelling and collision avoidance using Clifford algebra*. Lecture Notes in Computer Science **970** (1995), 699-704.
- [42] E. Hitzer, *Euclidean geometric objects in the Clifford geometric algebra of Origin, 3-Space, Infinity*. Bulletin of the Belgian Mathematical Society - Simon Stevin **11**(5) (2005), 653-662.
- [43] L.W. Yuan, Z.Y. Yu, W. Luo, et al., *A 3D GIS spatial data model based on conformal geometric algebra*. SCI China Ser D-Earth SCI **54**(1) (2011), 101-112.
- [44] Z.Y. Yu, *Multidimensional unified GIS data model based on GA*. Thesis of Nanjing Normal University for Doctoral Degree, 2011.
- [45] A. Jadczyk, *Quantum fractals on n -spheres. Clifford algebra approach*. Advances in Applied Clifford Algebras **17**(2) (2007), 201-240.
- [46] E. Hitzer, B. Mawardi, *Clifford Fourier transform on multivector fields and uncertainty principles for dimensions $n=2 \pmod{4}$ and $n=3 \pmod{4}$* . Advances in Applied Clifford Algebras **18**(3-4) (2008), 715-736.
- [47] E. Bayro-Corrochano, N. Arana-Daniel, *Clifford support vector machines for classification, regression and recurrence*. IEEE Transactions on Neural Networks **21**(1) (2010), 1731-1746.
- [48] K. Rieger, K. Schlacher, J. Holl, *On the observability of discrete-time dynamic systems - A geometric approach*. Automatica **44**(8) (2008), 2057-2062.

- [49] H. Eckhard, *Angles between subspaces computed in Clifford algebra*. AIP Conference Proceedings of International Conference of Numerical Analysis and Applied Mathematics (2010), 1476-1479.
- [50] B. Jancewicz, *Multivectors and Clifford algebra in electrodynamics*. World Scientific, 1988.
- [51] W. Luo, *Characteristics analysis and motion expression of temporal-spatial field data based on geometric algebra*. Thesis of Nanjing Normal University for Master Degree, 2011.
- [52] J. Y. Zhang, *GIS network analysis based on Clifford algebra*. Thesis of Nanjing Normal University for Master Degree, 2010.
- [53] L. Yi, *Multi-dimensional unified algorithm and application of voronoi based on conformal geometric algebra*. Thesis of Nanjing Normal University for Master Degree, 2011.
- [54] D. Hildenbrand, J. Pitt, A. Koch, *Gaalop-high performance parallel computing based on conformal geometric algebra*. In E. Bayro-Corrochano and G. Scheuermann (eds.), *Geometric Algebra Computing for Engineering and Computer Science*. Springer Verlag, London, 2010.
- [55] D. Fontijne, *Gaigen 2: a geometric algebra implementation generator*. In: 5th International Conference on Generative Programming and Component Engineering, Budapest (2006), 141-150.
- [56] A. Gentile, S. Segreto, F. Sorbello, *CliffoSor: A parallel embedded architecture for geometric algebra and computer graphics*. Seventh International Workshop on Computer Architecture for Machine Perception (CAMP 2005). Palermo, Italy (2005), 90-95.
- [57] L.W. Yuan, G. N. Lü, W. Luo, et al., *Geometric algebra method for multidimensionally-unified GIS computation*. Chinese Science Bulletin **57**(7) (2012), 802-811.
- [58] Z.Y. Yu, L.W. Yuan, W. Luo, et al., *Design and implementation of GIS temporal spatial analysis system based on Clifford algebra*. Geomatics and Information Science of Wuhan University **36**(12) (2011), 1397-1401.
- [59] Z.Y. Yu, L. W. Yuan, W. Luo, et al., *Boundary restricted non-overlapping sphere tree for unified multidimensional solid object index*. Journal of Software (in press).
- [60] L. Yi, L. W. Yuan, Z. Y. Yu, et al., *Clifford algebra-based voronoi algorithm*. Geography and Geo-information Science **27**(5) (2011), 37-41.

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